Requirements

Group 9 - Inside No.9:

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Introduction / Justification

We first made notes of our key takeaways and overall impressions of the Product Brief. Based on these, we pitched and discussed some initial ideas:

- Is the game going to be 2D, or 3D?
- First-person or third-person?
- Will the game be viewed ‘side-on’?
- Map design ideas: (From Central Hall to Halifax college)

We also looked into existing games with similar elements [1], to get a better sense of the broad direction for our game.

Based on our interpretation of the Product Brief and Assessment paper, we produced a list of preliminary requirements and constraints, and some clarifications to ask our client in a customer meeting. We collaborated with the customer on our questions and ideas to derive a list of mandatory and additional user requirements, based around things the player/user should be able to do.
We broke these down into functional and quality requirements by imagining what the game system would have to do to fulfill the user requirements. This process was guided by Sommerville’s ‘Software Engineering’ [2].

After attending the requirements engineering lecture [3], we agreed on presenting our requirements in the given Requirements Table format. Because the team and the markers are familiar with this layout, it ensures clear communication of requirements.

To resolve decisions on conflicting requirements and different project direction ideas between members, we have used a democratic voting system.

Adjustments have been made to a few of our functional requirements as we got further in development.
These were minor and not expected to affect the quality of the game.
We based the changes on what was easiest for our team to implement. For example, we changed some of the FR_Inventory requirements from requiring the scroll wheel and right click to select and use items, to being controlled by the number keys.

User Requirements

ID	Description	Priority
UR_Start-Screen	Game opens on a start screen: Providing initial inventory selection and start button.	Shall
UR_World	Game world will resemble the UoY campus, and will have limits that are intuitive to the player.	Shall
UR_Goal	The goal for the player is to reach a particular area, within a limited time, and avoiding ending the run without the goal accomplished. The game will make sure the player knows this goal.	Shall
UR_Character_Attributes	Character avatar will represent an undergraduate student at the UoY. Player will have to manage the character's health and energy properties.	Shall
UR_Timer	A run of the game will last a maximum of 5 minutes. Time remaining at the end of a run, with the goal complete, will positively affect the score.	Shall
UR_Geese	Many geese, portrayed as hungry, are an obstacle to the player's goal.	Shall
UR_Items	Items are collectable and usable by the player to affect the game world and the geese. Item use that better suits the goal will positively affect the score.	Shall
UR_Events	Events are triggered by the player and cause fun interactions. Managing events correctly will increase the score.	Shall
UR_Score	Game tracks a numerical score to evaluate player performance. A higher score at the end of a successful run, with the goal complete, represents a better result for the player.	Shall
UR_Movement	The player can move around within the limits of the campus world.	Shall
UR_Pause	Game allows pause at any point in a run.	Shall
UR_OSes	Users can access the game on at least 3 operating systems.	Shall
UR_Audience	Game is intended to be fun, casual & appropriate for A-level & undergrad aged people.	Shall
UR_Accessibility	Game is visually legible and usable across different screen sizes, and for players with common visual impairments.	Should
UR_Geese_Variations	Geese come in 2 or more models. Distinct geese are more rare, do more damage, and cannot be dispersed as easily.	May
UR_Difficulty-Set	Can choose between at least 2 difficulty options on the start screen. Makes the geese more challenging to deal with and limits inventory space.	May

Functional Requirements

ID	Description	User Requirements
FR_Start-Button	Start button triggers the start of a 'run' of the game.	UR_Start-Screen
FR_Inventory_Initial	On the start screen, the player can choose initial inventory items.	UR_Start-Screen
FR_Goal_Reached	When the player enters the Halifax Dorm area, the main goal is accomplished and the current run ends.	UR_Goal
FR_Inventory_Hotbar	Inventory slots are always displayed at the top of the game screen. Item icons are displayed within slots while the player possesses them.	UR_Items
FR_Item_Use	Number keys use/activate an item in the corresponding inventory slot.	UR_Items
FR_Movement	WASD or arrow keys will control player character movement.	UR_Movement
FR_Movement_Treadmill	The environment will move in the opposite direction of user input. While the player remains at the centre.	UR_Movement

ID	Description	User Requirements
FR_Collision	Game shall prevent the player character from moving through static environment colliders (walls and other obstacles).	UR_World
FR_Health	Player has a health property that is always displayed and can be increased or decreased by enemies, items and events.	UR_Character_Attributes
FR_Health_Zero	Player with no health triggers the current run to end.	UR_Goal
FR_Energy	Player has an energy property that is always displayed and can be increased or decreased by player actions, items and events.	UR_Character_Attributes
FR_Timer_Display	Time remaining will always be displayed.	UR_Timer
FR_Timer_Count	Timer starts at 300 at the beginning of a run and decreases by 1 every real world second	UR_Timer
FR_Timer_Zero	Timer at 0 triggers the current run to end.	UR_Timer
FR_Events	Secondary objectives and hidden events / easter eggs can be triggered by the player.	UR_Events
FR_Score	Score value is always displayed during a run.	UR_Score
FR_Geese_Vision	Geese can ‘see’ the player within a defined range, calculated based on relative positions.	UR_Geese
FR_Geese_Idle	A goose that can’t see the player will move in a random pattern.	UR_Geese
FR_Geese_Active	A goose that can see the player will move towards the player.	UR_Geese
FR_Geese_Distracted	Geese can be affected by items and events putting them in a different state than ‘idle’ or ‘active’.	UR_Geese
FR_Geese_Audio	Depending on player proximity to geese, there will be “demanding” sounding geese noises.	UR_Geese
FR_Run-End	The end of a run triggers the final calculation of score.	UR_Score + UR_Goal

Quality Requirements

ID	Description	User Requirements	Fit Criteria
QR_OS_Support	Game functions the same on at least 3 OSes.	UR_OSes	All functional requirements pass acceptance tests identically on Windows, macOS, and Linux.
QR_Timing_Display	Input and game state changes feel instantaneous to players.	UR_Audiance	Changes to the game state/world are displayed on screen within 100ms.
QR_Refresh-Display	Display refresh rate doesn’t go below 30Hz	UR_Audiance	Minimum 30 FPS on standard undergrad laptop hardware.
QR_Maintainability	Game source code is modular and documented so future developers can extend functionality,		A developer newly familiar with the code can implement a new item or event within 2 hours, using the project documentation.
QR_Usability	Game controls, limits and objectives are intuitive to the player. In-game messages are understandable.	UR_Accessibility + UR_Audiance	At least 80% of first-time players can understand the main goal and controls within the first 30 seconds without external guidance.
QR_Screen-Size	System automatically scales its viewport	UR_Accessibility	100% of UI elements remain visible

ID	Description	User Requirements	Fit Criteria
	and UI elements to provide a full view of the game on different sized screens.		across standard resolutions with no input from the user.

References

[1] poncle. Vampire Survivors. (2022). Accessed: Oct. 20, 2026. [Video Game]. Available: https://vampire.survivors.wiki/

[2] I. Somerville. “Requirements Engineering” In Software Engineering, 10th Global ed. London, England: Pearson, 2015, pp. 102-107. [Online]. Available: https://eu.alma.exlibrisgroup.com/leganto/public/44YORK_INST/citation/67444944630001381?auth=SAML

[3] K. Barmpis. (2026). Requirements Engineering [PowerPoint slides]. Available: https://vle.york.ac.uk/ultra/courses/_118516_1/document/_6763653_1?view=content&state=view